Report on
LABEL SMOOTHING

1. The Pre-Existing Problem

- ~~When~~ Traditionally, we use soft-max & cross-entropy loss $\left(\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}\right)$, $\text{Loss} = \sum_{k=1}^{K} -y_k \log \hat{y}_k$

along with one-hot vectors i.e. ~~$\hat{y}_i$~~ y_i is 1 if that is the correct class, else y_i is 0.
This is what use for multi-class problems.

- Training with these one-hot vectors has an unintended effect, the $\hat{y}_i$ will never reach 1, so the logits will - of the true class - try to reach $+\infty$ as that is the only way for $\hat{y}_i$ to become 1. On the contrary, the logits of wrong class will try to reach $-\infty$, also these logits will be very different

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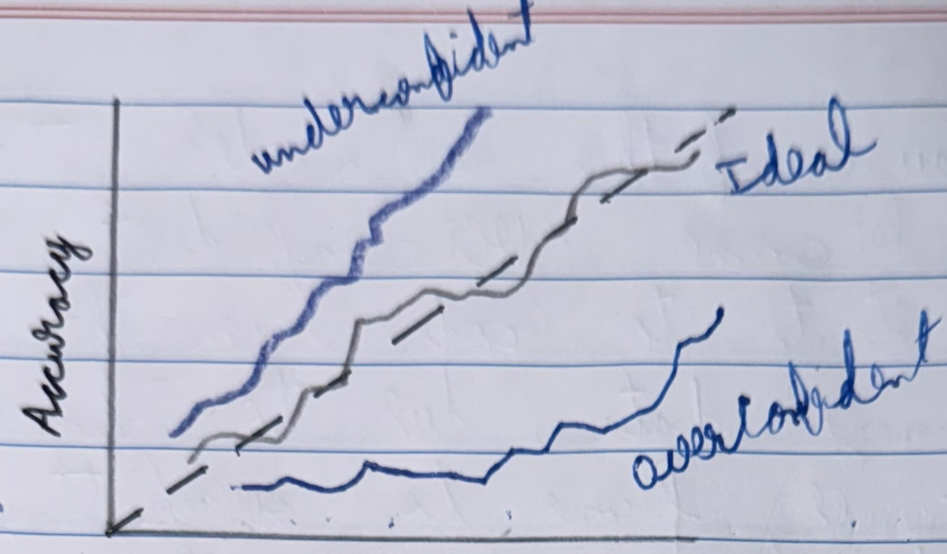
label smoothing which uses non-uniform distributions to better handle highly-imbalanced datasets.
There is also room in studying how label smoothing impacts knowledge distillation especially in complex sequence models & translation tasks.

a photo is 80% dog 15% cat & 5% car which means a dog ~~resembles~~ looks more like a cat than a car, whereas ~~in~~ a model which has been label smoothed might predict it is 80% dog 10% cat & 10% car reducing information. Similarly, it becomes a poor teacher for the same reason because it loses the required 'Dark Knowledge' to train a student network.

5. Future Improvements & Extensions

We can research the relationship b/w label smoothing & the information bottleneck principle, which studies how neural networks generalize by compressing data. Also future work may investigate unigram



3. What Label Smoothing is & how it works

- $\tilde{y}_i^{LS} = (1-\alpha)\, \tilde{y}_i + \dfrac{\alpha}{K}$

modified labels

α hyperparameter usually 0.1

Original labels

tot. no. of classes

what it does is we take a bit from the correct class & distribute it equally over the wrong classes.

eg. it was [1 0 0]

becomes [0.98 0.01 0.01]

- Because of this, our model starts to become overconfident as it starts to rote learn the training data, due to this it becomes prone to overfitting and poor generalization.

2. The Danger of Overconfident Predictions

An ideal Neural network should have its predicted probability almost equal to it's accuracy, this is called calibration. If predicted prob is more then ~~it~~ it's called overconfident eg. a class might be predicted w ~~100~~ 99 % confidence when it's right only 75% of the time. This overconfidence is detrimental for downstream Tasks such as beam-search algorithm in machine translation

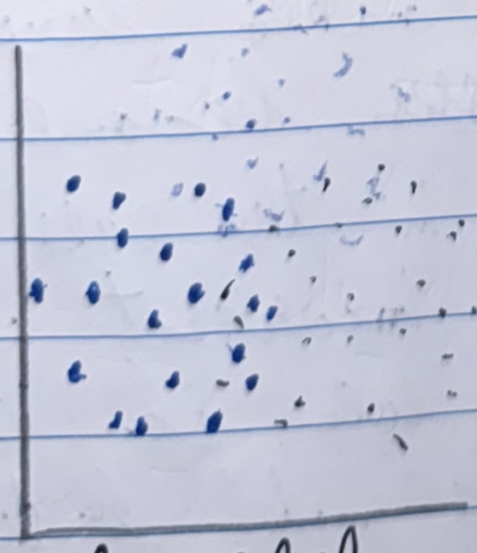
- $logit = x^T w_k$
  of kth class

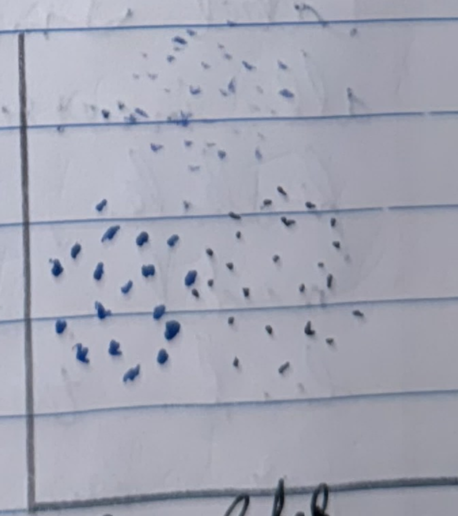
$\|x - w_k\|^2 = x^T x - 2x^T w_k + w_k^T w_k$

dist. from the correct class ∴ either factored out or constant across all classes

$\therefore \|x - w_k\|^2 \propto + x^T w_k$

So as the logits go towards ∞, so does the dist. & they start mixing with other classes.
On the contary, label smoothing sets a sort of maximum radius.





After label smoothing

4. Advantages & Limitations

Advantages:

- Label Smoothing configures the model automatically reducing ~~the~~ the overconfidence & thus reducing the need for post-processing ~~tt~~ Techniques like temperature Scaling. This helps in image classification & machine translation.

Limitations:

Label Smoothing makes sure that all incorrect classes become equally probale, so now in the eyes of the model, all the incorrect classes look the same, even if they are far from. eg. a model predicts that